PBL-I PROJECT REVIEW

Machine Learning – Unit 1 & Unit 2

Project Review Question Bank – 16 Groups

Course Portion	Unit 1 & Unit 2
PBL-I Review	20 Marks
Review Components	Problem Identification & Dataset Selection; EDA & Insights; Regression & Classification; Comparative Performance Analysis; Result Interpretation & Presentation
Number of Groups	16

Common Project Review Question Format

For the assigned project, answer the following five sections during the PBL-I Project Review. Each section carries 4 marks, for a total of 20 marks.

Q1. Problem Identification & Dataset Selection – 4 Marks

Define the real-world problem, formulate the machine-learning objective, identify the input and target variables, select an appropriate dataset, and justify why the selected dataset is suitable for the problem.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Perform data cleaning and exploratory analysis. Report missing values, descriptive statistics, mean, variance/standard deviation, quantiles, correlations/covariance where appropriate, distributions, outliers, and at least three meaningful observations from the dataset.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable regression and classification models from Unit 2. Use appropriate preprocessing and train/test splitting. Explain the selected algorithms and report the relevant performance measures.

Q4. Comparative Performance Analysis – 4 Marks

Compare the implemented models using appropriate evaluation metrics. For regression, consider MAE, MSE/RMSE and R^2. For classification, consider accuracy, precision, recall, F1-score and confusion matrix. Explain the observed differences in performance.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the results in the context of the problem. Identify the important findings, discuss model limitations, state the practical significance of the results, and present suitable graphs/tables and conclusions.

Group 1: House Price Prediction and Property Category Classification

Project Problem: Develop an ML system to predict house prices and classify properties into low-, medium-, and high-price categories using property and location attributes.

Suggested Dataset/Data: House-price dataset containing area, bedrooms, bathrooms, location, age, amenities, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of house price prediction and property category classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 2: Student Academic Performance Prediction and Classification

Project Problem: Develop an ML system to predict students' final marks and classify students according to academic performance using academic and attendance-related attributes.

Suggested Dataset/Data: Student dataset containing study hours, attendance, assignment marks, previous marks, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of student academic performance prediction and classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 3: Crop Yield Prediction and Productivity Classification

Project Problem: Develop an ML system to predict crop yield and classify agricultural productivity into low, medium, and high categories using environmental and farming parameters.

Suggested Dataset/Data: Agricultural dataset containing rainfall, temperature, soil nutrients, irrigation, crop type, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of crop yield prediction and productivity classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 4: Used Car Price Prediction and Condition Classification

Project Problem: Develop an ML system to predict the resale price of used cars and classify vehicles according to their condition/value category.

Suggested Dataset/Data: Used-car dataset containing year, mileage, engine size, fuel type, transmission, price, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of used car price prediction and condition classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 5: Electricity Consumption Prediction and Usage Classification

Project Problem: Develop an ML system to predict electricity consumption and classify consumption levels as low, medium, or high.

Suggested Dataset/Data: Electricity-consumption dataset containing time, consumption, temperature, humidity, appliance/load information, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of electricity consumption prediction and usage classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 6: Weather Temperature Prediction and Weather Condition Classification

Project Problem: Develop an ML system to predict temperature and classify weather observations into appropriate weather-condition categories.

Suggested Dataset/Data: Weather dataset containing temperature, humidity, pressure, wind speed, rainfall, cloud cover, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of weather temperature prediction and weather condition classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 7: Medical Cost Prediction and Healthcare Risk Classification

Project Problem: Develop an ML system to predict medical/insurance expenditure and classify individuals into suitable healthcare-risk categories.

Suggested Dataset/Data: Medical-cost dataset containing age, BMI, smoking status, children, region, medical expenses, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of medical cost prediction and healthcare risk classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 8: Employee Salary Prediction and Salary-Level Classification

Project Problem: Develop an ML system to predict employee salary and classify employees into low-, medium-, and high-salary levels using professional attributes.

Suggested Dataset/Data: Employee dataset containing experience, education, job role, age, skills, location, salary, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of employee salary prediction and salary-level classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 9: Student Placement Salary Prediction and Placement Classification

Project Problem: Develop an ML system to predict placement salary and classify students according to placement outcome using academic and skill-related attributes.

Suggested Dataset/Data: Placement dataset containing CGPA, internships, projects, skills, communication score, placement status, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of student placement salary prediction and placement classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 10: Rainfall Prediction and Rain/No-Rain Classification

Project Problem: Develop an ML system to predict rainfall amount and classify whether rainfall will occur for a given observation.

Suggested Dataset/Data: Weather/rainfall dataset containing temperature, humidity, pressure, wind speed, historical rainfall, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of rainfall prediction and rain/no-rain classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 11: Air Quality Prediction and Pollution-Level Classification

Project Problem: Develop an ML system to predict Air Quality Index (AQI) and classify air quality into suitable pollution-level categories.

Suggested Dataset/Data: Air-quality dataset containing PM2.5, PM10, NO2, SO2, CO, temperature, humidity, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of air quality prediction and pollution-level classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 12: Solar Power Generation Prediction and Generation-Level Classification

Project Problem: Develop an ML system to predict solar power generation and classify generation into low-, medium-, and high-generation levels.

Suggested Dataset/Data: Solar-energy dataset containing irradiance, temperature, humidity, panel parameters, time, power output, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of solar power generation prediction and generation-level classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 13: Customer Spending Prediction and Customer Segment Classification

Project Problem: Develop an ML system to predict customer spending and classify customers into meaningful spending or customer-segment categories.

Suggested Dataset/Data: Customer dataset containing age, income, purchases, visit frequency, spending score, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of customer spending prediction and customer segment classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 14: Telecommunication Customer Churn Prediction and Usage Analysis

Project Problem: Develop an ML system to predict customer expenditure/usage and classify customers according to churn status.

Suggested Dataset/Data: Telecommunication dataset containing call duration, data usage, recharge, tenure, complaints, plan type, churn, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of telecommunication customer churn prediction and usage analysis. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 15: Traffic Flow Prediction and Congestion Classification

Project Problem: Develop an ML system to predict traffic volume and classify traffic conditions into low-, medium-, and high-congestion levels.

Suggested Dataset/Data: Traffic dataset containing vehicle count, time, weather, road type, speed, occupancy, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of traffic flow prediction and congestion classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**

Group 16: Diabetes-Related Prediction and Patient Risk Classification

Project Problem: Develop an ML system to predict a suitable continuous health-related variable from the dataset and classify individuals according to diabetes outcome/risk.

Suggested Dataset/Data: Diabetes dataset containing glucose, BMI, age, blood pressure, insulin, pregnancies, diabetes outcome, etc.

Q1. Problem Identification & Dataset Selection – 4 Marks

Identify the variables required to solve the problem of diabetes-related prediction and patient risk classification. Select a suitable dataset and clearly define the input features, regression target and classification target. Justify the relevance, size and quality of the selected dataset.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Clean and explore the selected dataset. Calculate appropriate descriptive statistics including mean, variance/standard deviation and quantiles; examine distributions, missing values, outliers and relationships among variables; and report at least three useful insights.

Q3. Regression & Classification Implementation – 4 Marks

Implement suitable models from Unit 2 for both the continuous prediction and classification tasks. Compare appropriate models such as Linear Regression, Polynomial Regression, Ridge/Bayesian Regression for regression and Logistic Regression, Kernel methods and/or SVM for classification. Explain the preprocessing and model-training procedure.

Q4. Comparative Performance Analysis – 4 Marks

Evaluate and compare the implemented models. Use MAE, MSE/RMSE and R^2 for regression and accuracy, precision, recall, F1-score and confusion matrix for classification, as applicable. Identify the differences in model performance and explain possible reasons.

Q5. Result Interpretation & Presentation – 4 Marks

Interpret the obtained results for the selected real-world problem. Present suitable plots/tables, identify the major findings, discuss limitations and practical implications, and state possible future improvements.

Review Component	**Marks**
Problem Identification & Dataset Selection	4
Exploratory Data Analysis & Insights	4
Regression & Classification Implementation	4
Comparative Performance Analysis	4
Result Interpretation & Presentation	4
Total	**20**